

■ Excel Formula Cheat Sheet for Big Data Handling

■ Basic Formulas (Foundations)

Function	Formula	Description
SUM	=SUM(A1:A1000)	Adds numbers
AVERAGE	=AVERAGE(B2:B5000)	Finds mean
COUNT	=COUNT(C1:C10000)	Counts numeric entries
COUNTA	=COUNTA(D1:D10000)	Counts non-empty cells
MAX / MIN	=MAX(E:E) / =MIN(E:E)	Largest/smallest values
ROUND	=ROUND(F2,2)	Round to 2 decimals

■ Text Functions (Handling Unstructured Data)

Function	Formula	Description
LEN	=LEN(A2)	Counts characters
TRIM	=TRIM(B2)	Removes extra spaces
UPPER / LOWER	=UPPER(C2)	Change case
LEFT / RIGHT / MID	=LEFT(D2,5)	Extracts text parts
SEARCH / FIND	=SEARCH("Data",E2)	Finds text position
SUBSTITUTE	=SUBSTITUTE(F2,"old","new")	Replace text
TEXTJOIN	=TEXTJOIN(",",TRUE,A2:A10)	Combine values

■ Logical & Conditional

Function	Formula	Description
IF	=IF(A2>100,"High","Low")	Conditional logic
AND / OR	=IF(AND(B2>50,C2<100),"Yes","No")	Multiple conditions
IFERROR	=IFERROR(A2/B2,"Error")	Handle errors

■ Lookup & Reference

Function	Formula	Description
VLOOKUP	=VLOOKUP(123, A2:D10000, 2, FALSE)	Search vertically
INDEX + MATCH	=INDEX(C2:C10000, MATCH(500,A2:A10000,0))	Flexible lookup
XLOOKUP	=XLOOKUP("ID123", A:A, B:B)	Modern lookup
INDIRECT	=INDIRECT("Sheet2!A1")	Dynamic reference

■ Date & Time

Function	Formula	Description
TODAY / NOW	=TODAY()	Current date
DATEDIF	=DATEDIF(A2,B2,"D")	Days between dates
TEXT	=TEXT(Date,"yyyy-mm")	Format dates
NETWORKDAYS	=NETWORKDAYS(A2,B2)	Working days count

■ Statistical & Data Analysis

Function	Formula	Description
STDEV.S / STDEV.P	=STDEV.S(A:A)	Standard deviation
VAR.S / VAR.P	=VAR.S(A:A)	Variance
CORREL	=CORREL(A2:A1000,B2:B1000)	Correlation
PERCENTILE.INC	=PERCENTILE.INC(C2:C1000,0.9)	90th percentile
RANK.EQ	=RANK.EQ(D2,D2:D1000)	Ranking values

■ Data Cleaning & Transformation

Function	Formula	Description
UNIQUE	=UNIQUE(A2:A10000)	Distinct values
FILTER	=FILTER(A2:C10000,B2:B10000="Yes")	Filter rows
SORT	=SORT(A2:C10000,2,-1)	Sort data

■ Advanced & Array Formulas

Function	Formula	Description
SUMPRODUCT	=SUMPRODUCT(A2:A1000,B2:B1000)	Weighted sum
MMULT	=MMULT(A1:B2,C1:D2)	Matrix multiplication
TRANSPOSE	=TRANSPOSE(A1:B5)	Convert rows/columns